Subgraph relevance / role judge

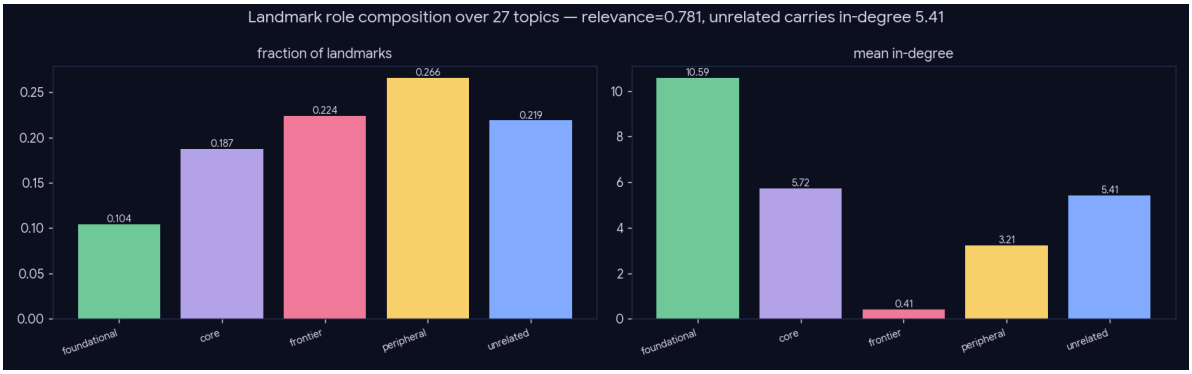
arxivkg · retrieval · precision + composition, gold-free

What we did

Where `02-01` measures *recall* against curated gold (3 topics), this measures the two things recall cannot: **precision** (how much of the retrieved subgraph is off-topic noise) and **composition** (does it span a field's lineage). It needs no gold, so it runs the whole 27-topic battery. For every committed landmark (induce/PPR), a self-hosted Gemma-4-26B judge (@T= 0) sees *only* the topic + the paper's title/year/abstract — not our PPR/in-degree scores, so its label is an independent check — and assigns one role: `foundational`, `core`, `frontier`, `peripheral`, `unrelated`. Derived: `relevance_rate` = 1− unrelated-frac; `lineage_span` ($\geq$ 1 foundational AND $\geq$ 1 frontier); and mean in-degree per role (a structure-agreement check vs PPR).

Results (27 topics, 1080 landmarks)

role	fraction	mean in-degree
foundational	0.104	10.59
core	0.187	5.72
frontier	0.224	0.41
peripheral	0.266	3.21
unrelated	0.219	5.41



Conclusion

~78% of retrieved landmarks are on-topic (relevance = 0.781**), and 0.889 of topics span their full lineage** — compositionally complete enough to ground a review. **But ~22% is off-topic noise (0.219), and it is structurally invisible to PPR**: unrelated papers carry mean in-degree 5.41 — as high as `core` — because they are famous cross-field hubs (GPT-3 / StyleGAN under "diffusion") that ride in on citation fame, not relevance. PPR cannot tell them from genuine core work; only the content judge can. This is the precise failure `02-03` targets: the noise enters

through the most-cited-ancestor injection, and a cosine floor removes it. **Caveat:** the role judge is the measurement instrument; its fairness is validated separately (judge-validity, in `03-generation`).